

Oseledec's theorem deals with the asymptotic behavior of infinitesimal linear perturbations along orbits of a general nonlinear system. We will not study it in its full generality here. Rather, we will only look at its major consequences: definition of Lyapunov exponents and associated subspaces (called Oseledec spaces) of tangent spaces. We will define chaotic systems from there.

Let's introduce Oseledec's multiplicative ergodic theorem as applied to linearizations of discrete-time dynamical systems or maps

$F: M \rightarrow M$ M compact subset of $\mathbb{R}^n$
(nonlinear map)

$$\nabla F: TM \rightarrow TM$$

Remember the linearization

$$v_{t+1} = \nabla F(x_t) v_t$$

$$v_t \in \mathbb{R}^n$$

is an infinitesimal perturbation at point x_t along a fixed reference orbit $\{x_t\}$.

Let $A_t \equiv \nabla F(x_t)$ be the sequence of Jacobian matrices.

When A_t is some fixed matrix A , as happens when you take $x_t = x^*$, a fixed point, we already know the solutions $v_t = e^{tA} v_0$.

But what happens along more general orbits? In particular, aperiodic orbits?

For linearizations of periodic orbits, see lecture 12 notes on the monodromy matrix

Along a generic orbit, to understand the behavior of linearization, Oseledec's multiplicative ergodic theorem comes to our rescue.

$$\begin{array}{c} v_t \\ \uparrow \\ \text{tangent} \\ \text{vector at } x_t \end{array} = \begin{array}{c} A_{t-1} \cdots A_0 v_0 \\ \downarrow \\ DF(x_{t-1}) \end{array}$$

Let the Jacobian matrices $A_t = DF(x_t)$ be nonsingular $n \times n$
Let us define $A(t) = A_{t-1} \cdots A_0 \in \mathbb{R}^{n \times n}$

OMET says that as long as $\frac{1}{t} \log \|A(t)\| \rightarrow 0$

ie the Jacobians do not increase or decrease on an exponential scale as $t \rightarrow \infty$, then for any nonzero $v_0 \in \mathbb{R}^n$, the following limit exists:

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \|A(t) v_0\|$$

Note that this limit depends on two things: $\{x_t\}$ (and hence x_0) and v_0 .

This limit is called the characteristic exponent or Lyapunov exponent:

$$\lambda(x_0, v_0) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \|A(t) v_0\|$$

In words, this is the rate of asymptotic, exponential growth or decay rate of

infinitesimal perturbations along the orbit $\{x_t\}$ if the initial infinitesimal perturbation is v_0 at x_0 .

Now, OMET says that Lyapunov exponents exist and they take on a finite number of values, usually $p \leq n$ distinct values, at each x_0 & v_0 . usually arranged in decreasing order $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$. In a chaotic system, the top LE $\lambda_1(x_0, v_0) > 0$ for almost every vector $v_0 \in \mathbb{R}^n$.

That is, almost every infinitesimal perturbation grows exponentially in a chaotic orbit.

This gives the appearance of "randomness" to a chaotic orbit because when small perturbations can cause nearby orbits to become uncorrelated, it becomes difficult to predict the exact evolution of any chaotic orbit, even though the dynamics is deterministic.

OMET also says that the sequence $\{A(t)\}$ is Lyapunov regular. That is, the following limit exists and is equal to sum of LEs with their respective multiplicities.

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log |\det A(t)| = \sum_{i=1}^p k_i \lambda_i \rightarrow \textcircled{A}$$

where k_i is the "multiplicity" of λ_i , which we will discuss below.

First, let's understand what the determinant does.

The matrix $A(t)$ (really any matrix) is a linear operator on vectors in $\mathbb{R}^n$. The $|\det A(t)|$ is the ratio of volumes before after to before applying this linear operator.

So this statement is saying that the factor by which volumes are expanded / contracted on a exponential scale is given by the sum of LEs.

Let's take a one-dimensional subspace of $\mathbb{R}^n$ - a line = $\text{span}\{v_0\}$

The amount of expansion / contraction on an exponential scale is $\frac{1}{t} \log \|A(t)v_0\|$, which converges to

an LE as $t \rightarrow \infty$.

Now take a b -dimensional volume, $b \geq 1$.

$$\text{This amount } \text{vol}_b(t) = \prod_{i=1}^b (\text{sing. value}_i \text{ of } A(t))$$

$$\Rightarrow \frac{1}{t} \log \text{vol}_b(t) = \frac{1}{t} \sum_{i=1}^b \log(\text{sing. value}_i \text{ of } A(t))$$

which converges to the sum of the top b LEs.

OMET says that one can split $\mathbb{R}^n$ into subspaces characterized by the LEs.

$$\text{Let's say } V_1 = \{v_0 : \lambda(x_0, v_0) \leq \lambda_1(x_0)\}$$

Clearly, $V_1 \equiv \mathbb{R}^n$ at every x_0 .

$$\text{Similarly, let } V_2 = \{v_0 \in \mathbb{R}^n : \lambda(x_0, v_0) \leq \lambda_2(x_0)\}$$

Clearly, $V_2 \subset V_1$

And like this, if p is the number of distinct LEs, then, we have the tangent vectors at x_0

having an ~~decomposition~~ ordering (also called a filtration)
 as $V_1 \equiv \mathbb{R}^n \supset V_2 \supset V_3 \dots \supset V_p \supset V_{p+1} = \{0\}$

With this one can also tell exactly what subspace
 of $\mathbb{R}^n$ will produce an exponential growth/decay
 at exactly λ_i . This is $V_i \setminus V_{i+1}$.

The dimension of $V_i \setminus V_{i+1}$ (V_i but not V_{i+1})
 = k_i , called multiplicity of λ_i .

For a chaotic system, $V_1 \setminus V_{p+1}$ has dimension
 at least 1. That is, there exists a non-empty
 unstable subspace (corresponding to $\lambda_1 > 0$).
 at any x

Although we have discussed linearizations under
 discrete-time dynamics / maps, OMET equally applies
 to linearizations of flows.

$$\frac{\partial \phi^t(x_0)}{\partial t} = V(\phi^t(x_0)) \quad \text{linearization:} \quad \frac{d u(t)}{dt} = DV(\phi^t(x_0)) u(t)$$

Again let $A_t = DV(\phi^t(x_0)) = DV(x_t)$
 although now $t \in \mathbb{R}$ as opposed to an integer

Since the linearization is a linear (albeit nonautonomous) ODE, we have

$$u(t) = F(t) u(0)$$

where $F(t) \in \mathbb{R}^{n \times n}$ is a matrix called the
 tangent propagator.

Ex: what is the connection b/w $F(t)$ and the monodromy matrix when $x(t)$ is periodic?

We have the same consequence of Lyapunov regularity:

$$\lim_{t \rightarrow \infty} (F(t)^T F(t))^{\frac{1}{2t}} = \Lambda$$

where Λ is a diagonal matrix with entries e^{λ_i}

and the λ_i & V_i are defined in the same way.

Ex: Show that $\lambda_i = 0$ is a valid LE for a flow with non-singular A_t 's and the corresponding $V_i | V_{\perp}$ (Oseledec subspace) is at least one-dimensional.

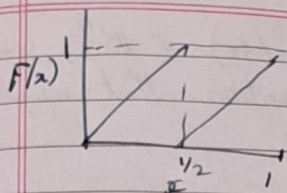
Hint: Consider what happens to the tangent equation with $u(0) = V(x_0)$.

chaotic system examples

Let's consider this 1D map, called the Sauterth map:

$$F(x) = 2x \text{ } \cdot\!\! \cdot\!\! \cdot\!\! 1 \quad \cdot\!\! \cdot\!\! \cdot\!\! : \text{ modulus operation}$$

$$= \begin{cases} 2x & x \leq \frac{1}{2} \\ 2x-1 & x > \frac{1}{2} \end{cases}$$



For x_0 being a dyadic number
(representable on a computer)

$F^n(x_0) = 0$ for some large
finite ~~enough~~ $n = 0$.

Ex: Show that there are $2^n - 1$ periodic points
of period n .

On a computer, since this is just the left shift
operator applied to a number in binary, every
orbit eventually goes to 0. To simulate the
"typical" behavior, we must add a small noise,
i.e. $F(x) = (2x + \epsilon) \cdot 1$

"Typical" behavior is what happens to almost
every initial condition, i.e. irrational points on
 $[0,1]$. — these form chaotic orbits.

Check that the Lyapunov exponent for almost every
 x_0 is $\log 2$.

Every perturbation is unstable!

In higher dimensions, we may also have stable
subspaces, i.e. $A(t)$ does not need to ~~contract~~ ^{expand}
volumes in every subspace in $\mathbb{R}^n$ even in a chaotic
orbit. As we know from previous lectures,

$$E^u(x_0) = \{v_0 : \lambda(x_0, v_0) > 0\}$$

$$E^s(x_0) = \{v_0 : \lambda(x_0, v_0) < 0\}$$

Note also that we can define E^u & E^s at each
point (they are different subspaces at each x_0)
in the same way.

The decomposition $E^u(x_0) \oplus E^s(x_0) = \mathbb{R}^n$, as before is a direct sum decomposition that applies to hyperbolic maps.

In hyperbolic flows, we also have a center direction

$$E^c(x_0) = \{v_0 : \lambda(x_0, v_0) = 0\}$$

So, $E^u(x_0) \oplus E^s(x_0) \oplus E^c(x_0) = \mathbb{R}^n$ at each x_0 .

Now, what is the connection b/w this direct sum decomposition and the Oseledec subspaces we discussed before?

To understand this consider the singular value decomposition

$$F(t) = U(t) \Sigma(t) W^T(t)$$

(or $A(t)$ in the case of maps)

$$\text{where } U(t) = [q_1(t) | \dots | q_n(t)]$$

consists of the left singular vectors,

$$W(t) = [w_1(t) | \dots | w_n(t)]$$

consists of the right singular vectors

$$\text{and } \Sigma(t) = \begin{bmatrix} \sigma_1(t) & & \\ & \ddots & \\ & & \sigma_n(t) \end{bmatrix}$$

There are other bases for V_i that are easily interpretable and we will not discuss them in class or in the homework.

The code in PS4 computes ^{numerically} a different basis of backward Lyapunov vectors.

In summary, using linear algebraic operations, like taking the SVD, we can approximate Oseledets spaces that have different exponential growth/decay rates.

In this way, we can understand E^u & E^s & E^c even in high-dimensional chaotic systems. But first, let's look at other 2D map examples: Smale's horseshoe map and the Baker's map.

The Baker's map & Smale's horseshoe map are topologically conjugate, i.e. you can map the orbits of one to the other through a continuous coordinate transformation.

Ex: Simulate the Baker's map on a computer.

For these maps, the unstable manifold is dense (the global unstable manifold of a typical orbit) is $[0, 1]^2$.

The map is volume preserving.

Ex: Show that (using the change of variables in integration that for any function f

$$\int_{x \sim \mu} f(x) = \int_{x \sim \mu} f(F(x))$$

consists of the singular values.

As we noted before, OMET says that $F(t)$ is Lyapunov regular, which means that the following limit exists and is a diagonal matrix

$$\lim_{t \rightarrow \infty} (F(t)(F(t))^T)^{\frac{1}{2t}} = \Lambda$$

with elements being e^{λ_i}

This means that, substituting the SVD, we get

$$\lim_{t \rightarrow \infty} W(t)(\Sigma(t))^{\frac{1}{2t}}W(t) = I$$

and as $t \rightarrow \infty$, $W(t)$ becomes independent t and only a function of x_0 .

$W(x_0)$ are the eigenvectors of $(F(t))^T F(t)$ as $t \rightarrow \infty$. So it is the and the eigenvalues converge to e^{λ_i} .

Also

If we only consider $\lambda_1, \lambda_2, \dots, \lambda_n$ where $\lambda_{n+1} < 0$ and $\lambda_n > 0$. Then,

the corresponding eigenvectors, also called (forward) Lyapunov vectors provide an orthonormal basis for the filtering $V_1 = \mathbb{R}^n \supset V_2 \supset \dots \supset V_p \supset V_p^\perp$ that we saw before.

Recall that $V_i = \{v_0 : d(x_0, v_0) \leq d_i\}$

Ex: show that $V_i = \text{span}\{w_j : j=i, i+1, \dots\}$

where μ is the uniform distribution on $[0,1]^2$ and F is the Baker's map, which is given by,

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{pmatrix} 2x - \lfloor 2x \rfloor \\ \frac{y + \lfloor 2x \rfloor}{2} \end{pmatrix}$$

The "x" component is the same map as the Sawtooth map but F here is invertible.

Ex. Identify the attractor (set that is forward and backward invariant under the dynamics that all orbits asymptotically enter) for Smale's horseshoe with

$$\Lambda = \left\{ x \in B \cap U D : F^t(x) \in B \cap U D \right. \\ \left. \forall t \in \mathbb{Z} \right\}$$

